



Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no.....

Name of the Course: MCA(AI&DS)
Semester: 2
Name of the Paper: Artificial Intelligence
Paper Code: TMD 201
Time: 1.5 hours

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub-questions
- (ii) Each question carries 10 marks.

- Q1. (10 Marks)
- a. Differentiate between an Agent program and an Agent function, taking a vacuum cleaning agent as an example. Is the input to an agent program the same as the input to the agent function? [CO1]
- OR**
- b. An online tutoring system adapts lessons based on a student's performance, providing additional practice on weak topics and advancing faster on strong areas. Explain the type of agent? What are its key features? [CO1]
- Q2. (10 Marks)
- a. AI-based proctoring system monitors students during online examinations using webcam video, audio, and screen activity to detect suspicious behavior. [CO1]
- 1. Specify the PEAS components for the proctoring system.
 - 2. Determine the environment type.
 - 3. What agent type is most suitable for this application?
- OR**
- b. Explain Hill climbing algorithm, focusing on the major challenges of the algorithm [CO2]
- Q3. (10 Marks)
- a. Differentiate between Heuristic and Meta Heuristic algorithms with an example for each. [CO2]
- OR**
- b. In the context of a problem-solving agent, explain why goal formulation precedes problem formulation. [CO2]
- Q4. (10 Marks)
- a. Genetic algorithm is used to maximize the function given below,
 $f(x) = x^2$; x is in the range $0 < x < 31$
The binary chromosomes are given as: 01001; 11100; 00100; 11001
Calculate the fitness function and identify the best chromosome. [CO2]
- OR**
- b. Explain the PSO algorithm in detail. [CO2]

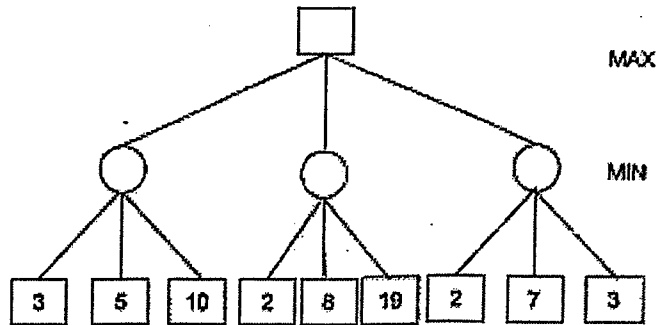


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Q5.

(10 Marks)

- a. Consider the game with 9 final states as shown above. As a maximizing player, which move would you make, considering that your opponent also plays optimally? [CO2]



OR

- b. Apply Alpha-Beta pruning to the given game tree below and determine the optimal moves and the nodes that are pruned. [CO2]

